Date: 21/01/26

# Program 1: **Lowest Product Price**

**Aim:** Define a class 'product' with data members pcode, pname and price. Create 3 objects of the class and find the product having the lowest price.

## Algorithm:

1. Create a class named **product.** Declare instance variables **pcode, pname** and **price.**
2. Declare a method **details** that takes a integer **n** as a parameter.
   - Prompt the user to enter product details for the n$^{th}$ item.
   - Read the input values and assign them to the respective instance variables.
3. Declare a method **lowest** that takes two **product** objects as parameters(**a** and **b**).
   - Compare the price of the current product object (**this)**, price of object **a,** and price of object **b**. Find the lowest price. ( if this.price < (a.price and b.price) then this.price is the lowest, else if a.price<b.price then a.price is the lowest, else b.price is the lowest).
   - Display the product having lowest price.
4. Instantiate three objects of **product** class (**p1, p2, p3**).
5. Call the **details** method for each object.
6. Call the **lowest** method with **p1** object, passing **p2** and **p3** as parameters.

## Source Code:

```
import java.util.Scanner;
class product
{
    String pcode,pname;
    int price;
    void details(int n)
    {
        Scanner sc=new Scanner(System.in);
        System.out.print("\nEnter product code for item "+n+": ");
        pcode=sc.next();
        System.out.print("Enter product name for item "+n+": ");
        pname=sc.next();
        System.out.print("Enter product price for item "+n+": ");
        price=sc.nextInt();
    }
    void lowest(product a,product b)
    {
        if(price<a.price && price<b.price)
            System.out.println("\nProduct "+pname+" has lowest price");
        else if(a.price<b.price)
            System.out.println("\nProduct "+a.pname+" has lowest
             price");
```

```
        else
            System.out.println("\nProduct "+b.pname+" has lowest
             price");
    }
}
class compare_objects
{
    public static void main(String[] args)
    {
        product p1=new product();
        product p2=new product();
        product p3=new product();
        p1.details(1);
        p2.details(2);
        p3.details(3);
        p1.lowest(p2, p3);
    }
}
```

## Output:

```
Enter product code for item 1: 001
Enter product name for item 1: ABC
Enter product price for item 1: 1200

Enter product code for item 2: 002
Enter product name for item 2: XYZ
Enter product price for item 2: 2500

Enter product code for item 3: 003
Enter product name for item 3: IJK
Enter product price for item 3: 1000

Product IJK has lowest price
```

**Result:** The program has executed successfully and required output is obtained.

Date: 21/01/26

# Program 2: **Matrix Addition**

**Aim:** Read 2 matrices from the console and perform matrix addition.

## **Algorithm:**

1. Create class named **addmatrix.**
    - Declare instance variables **n** and **m** for the size of the matrix, a 2D array **matrix** to store the matrix elements.
2. Declare a **constructor** where user is prompted to enter the size of the matrix.
    - Initialize the matrix array with the specified size.
    - Prompt the user to enter the values of the matrix elements.
3. Declare a method named **add** that takes another **addmatrix** object **b** as a parameter.
    - Check if the sizes of the two matrices are compatible for addition (number of rows and columns match).
    - If not compatible, print a message indicating that addition is not possible.
    - If compatible, perform element-wise addition of the matrices and update the current matrix.
    - Display the sum matrix by calling the display method.
4. Declare a method named **display** to print the elements of a matrix.
    - Use nested loops to iterate through the rows and columns of the matrix.
    - Print each element of the matrix followed by a space.
    - Print a newline character after each row.
5. Instantiate two objects of the **addmatrix** class (**a** and **b**).
6. Call the **add** method of the first matrix (**a**), passing the second matrix (**b**) as a parameter.

## **Source Code**:

```
import java.util.*;
class addmatrix
{
    Scanner sc=new Scanner (System.in);
    int n,m;
    int matrix[][];
    addmatrix(int x)
    {
        System.out.print("Enter size of matrix "+x+": ");
        n=sc.nextInt();
        m=sc.nextInt();
        matrix=new int[n][m];
        System.out.println("Enter value of matrix "+x+": ");
        for(int i=0;i<n;i++)
            for(int j=0;j<m;j++)
                matrix[i][j]=sc.nextInt();
    }
```

```
    void add(addmatrix b)
    {
        if(n!=b.n || m!=b.m)
            System.out.println("Addition of matrix not possible");
        else
        {
            for(int i=0;i<n;i++)
                for(int j=0;j<m;j++)
                matrix[i][j]+=b.matrix[i][j];
            System.out.println("Sum of matrix");
            this.display();
        }
    }
    void display()
    {
        for(int i=0;i<n;i++)
        {
            for(int j=0;j<m;j++)
                System.out.print(matrix[i][j]+" ");
            System.out.println();
        }
    }
}
class addmatrix_main
{
    public static void main(String[] args)
    {
        addmatrix a=new addmatrix(1);
        addmatrix b=new addmatrix(2);
        a.add(b);
    }
}
```

**Output:**

```
Enter size of matrix 1: 3 3
Enter value of matrix 1:
1 2 3
4 5 6
7 8 9
Enter size of matrix 2: 3 3
Enter value of matrix 2:
9 8 7
6 5 4
3 2 1
Sum of matrix
10 10 10
10 10 10
10 10 10
```

**Result:** The program has executed successfully and required output is obtained.

Date: 21/01/26

# Program 3: **Add 2 Complex Numbers**

**Aim:** Read 2 complex numbers and perform addition on the 2 complex numbers.

## Algorithm:

1. Create class named **complex**.
   - Declare instance variables **x** and **y** to represent the real and imaginary parts of a complex number.
2. Declare a method named **input** that takes an integer n as a parameter.
   - Prompt the user to enter the real and imaginary parts of the complex number for the $n^{th}$ instance.
   - Read the input values and assign them to the instance variables **x** and **y.**
3. Declare a method named **add** that takes another **complex** object **b** as a parameter.
   - Create a new **complex** object **c** to store the result of the addition.
   - Calculate the sum of the real parts (**x**) and imaginary parts (**y**) of the current object (**this**) and the object **b**.
   - Display the result of the addition in the form of complex numbers.
4. Instantiate two objects of the **complex** class (**a** and **b**).
5. Call the **input** method for each object to input the real and imaginary parts.
6. Call the **add** method of the first complex number (**a**), passing the second complex number (**b**) as a parameter.

## Source Code:

```
import java.util.*;
class complex
{
    int x,y;
    Scanner sc=new Scanner(System.in);
    void input(int n)
    {
        System.out.print("\nEnter real part of the number "+n+": ");
        x=sc.nextInt();
        System.out.print("Enter imaginary part of the number "+n+": ");
        y=sc.nextInt();
    }
    void add(complex b)
    {
        complex c=new complex();
        c.x=x+b.x;
        c.y=y+b.y;
        System.out.println("\n("+x+" +i"+y+") + ("+b.x+" +i"+b.y+") =
         "+c.x+" +i"+c.y);
    }
```

```
}
class complex_main
{
    public static void main(String[] args)
    {
        complex a=new complex();
        complex b=new complex();
        a.input(1);
        b.input(2);
        a.add(b);
    }
}
```

**Output:**

```
Enter real part of the number 1: 2
Enter imaginary part of the number 1: 3

Enter real part of the number 2: 4
Enter imaginary part of the number 2: 5

(2 +i3) + (4 +i5) = 6 +i8
```

**Result:** The program has executed successfully and required output is obtained.

Date: 28/01/26

# Program 4: **Symmetric Matrix**

**Aim:** Read a matrix from the console and check whether it is symmetric or not.

## Algorithm:

1. Create class named **symmetric**.
   - Declare instance variables **n** and **m** to represent the size of the matrix,
   - a 2D array **matrix** to store the matrix elements,
   - a variable **c** to track whether the matrix is symmetric (initialized to 0).
2. Define a constructor which prompts the user to enter the size of the matrix
   - Initialize the **matrix** array with the specified size.
   - Prompt the user to enter the values of the matrix elements.
3. Declare a method named **transpose** to check if the matrix is symmetric.
   - Use nested loops to iterate through the rows and columns of the matrix.
   - Check if the element at position **(i, j)** is equal to the element at position **(j, i)** for all **i** and **j**.
   - If any such pair of elements is not equal, set **c** to 1 and break out of the loop.
   - If **c** remains 0 after the loops, print that the matrix is symmetric; otherwise, print that it is not symmetric.
4. Instantiate an object of the **symmetric** class (**a**).
5. Call the **transpose** method to check if the matrix is symmetric.

## Source Code:

```
import java.util.*;
class symmetric
{
    Scanner sc=new Scanner (System.in);
    int n,m,matrix[][];
    int c=0;
    symmetric()
    {
        System.out.print("Enter size of matrix: ");
        n=sc.nextInt();
        m=sc.nextInt();
        matrix=new int[n][m];
        System.out.println("Enter value of matrix: ");
        for(int i=0;i<n;i++)
            for(int j=0;j<m;j++)
            matrix[i][j]=sc.nextInt();
    }
    void transpose()
    {
        for(int i=0;i<n;i++)
```

```
        {
            for(int j=0;j<m;j++)
                if(matrix[j][i]!=matrix[i][j])
                {
                    c=1;
                    break;
                }
        }
        if(c!=1)
            System.out.println("Matrix is symmetric");
        else
            System.out.println("Matrix is not symmetric");
    }
}
class symmetric_main
{
    public static void main(String[] args)
    {
        symmetric a=new symmetric();
        a.transpose();
    }
}
```

**Output:**

```
Enter size of matrix: 3 3
Enter value of matrix:
1 2 3
2 0 4
3 4 5
Matrix is symmetric
```

**Result:** The program has executed successfully and required output is obtained.

Date: 28/01/26

# Program 5: **Inner Class**

**Aim:** Create CPU with attribute price. Create inner class Processor (no. of cores, manufacturer) and static nested class RAM (memory, manufacturer). Create an object of CPU and print information of Processor and RAM.

## Algorithm:

1. Create a class named **CPU.** Declare instance variables **price.**
2. Define a non-static inner class **Processor** within the **CPU** class:
    - Declare integer variable **no_cores** and string variable **manufacturer**.
    - In the constructor of this class, read and set **no_cores, manufacturer.**
3. Define a static inner class **RAM** within the **CPU** class:
    - Declare integer variable **memory** and string variable **manufacturer**.
    - In the constructor of this class, read and set **memory, manufacturer.**
4. Create an instance of the **CPU** class (**c**).
5. Create an instance of the non-static inner class **Processor** using the object of CPU class.
6. Create an instance of the static inner class **RAM**.
7. Calculate the CPU price based on the no: of cores in the processor and the RAM capacity.
8. Print the details of the processor and RAM using the information obtained from the user input.

## Source Code:

```
import java .util.*;
class CPU
{
    Scanner sc= new Scanner(System.in);
    int price;
    class Processor
    {
        int no_cores;
        String manufacturer;
        Processor()
        {
            System.out.print("Enter no:of cores in the processor: ");
            no_cores =sc.nextInt();
            System.out.print("Enter manufacturer of the processor: ");
            manufacturer =sc.next();
        }
    }
    static class RAM
    {
        Scanner sc= new Scanner(System.in);
        int memory;
        String manufacturer;
```

```
        RAM()
        {
            System.out.print("Enter memory capacity of RAM: ");
            memory =sc.nextInt();
            System.out.print("Enter manufacturer of the RAM: ");
            manufacturer =sc.next();
        }
    }
}
class innerclass
{
    public static void main(String[] args)
    {
        CPU c = new CPU();
        CPU.Processor p=c.new Processor();
        CPU.RAM r= new CPU.RAM();
        c.price=2500+(p.no_cores*20)+(r.memory*15);
        System.out.println("\n Processor:\nNo:of Cores :
         "+p.no_cores+"\nManufacturer: "+p.manufacturer);
        System.out.println("\n RAM:\nMemory Capacity : "+r.memory+"
         GB\nManufacturer: "+r.manufacturer);
        System.out.println("\nCPU Price: "+c.price);
    }
}
```

## <u>Output:</u>

```
Enter no:of cores in the processor: 8
Enter manufacturer of the processor: Intel
Enter memory capacity of RAM: 16
Enter manufacturer of the RAM: Corsair

 Processor:
No:of Cores : 8
Manufacturer: Intel

 RAM:
Memory Capacity : 16 GB
Manufacturer: Corsair

CPU Price: 2900
```

**Result:** The program has executed successfully and required output is obtained.

Date: 28/01/26

# Program 6: **Search Element In Array**

**Aim:** Program to search for an element in an array.

## Algorithm:

1. Create a class named **array.** Declare instance variables **arr** (an integer array), **size** (to store the size of the array).
2. Define a constructor that takes an integer **n** as a parameter.
   - Initialize **size** with **n** and create an integer array **arr** of size **n**.
   - Prompt the user to enter the elements to array (**arr**).
3. Define a method named **search** that takes an integer **n** as a parameter.
4. Initialize a variable **c** to 0 to represent whether the number is found in the array.
5. Iterate through the array and check if the element at the current index is equal to **n**.
   - If found, set **c** to 1, print that the number is found, and break out of the loop.
   - If not found, print that the number is not an element of the array
6. In the **main** method, Prompt the user to enter the number of elements in the array (**n**).
   - Create an object **a** of the class **array.**
   - Prompt the user to enter the number to be searched (**num**).
   - Call the **search** method of the **array** object (**a**) with the specified number (**num**).

## Source Code:

```
import java.util.*;
class array
{
    Scanner sc=new Scanner(System.in);
    int arr[],size;
    array(int n)
    {
        size=n;
        arr=new int[n];
        System.out.println("Enter elements to the array: ");
        for(int i=0;i<n;i++)
            arr[i]=sc.nextInt();
    }
    void search(int n)
    {
        int c=0;
        for(int i=0;i<size;i++)
            if(arr[i]==n)
            {
                c=1;
                System.out.println(n+" is found in the array.");
                break;
```

```
                }
            if(c==0)
                System.out.println(n+" is not an element of the array.");
        }
    }
    class search_array
    {
        public static void main(String[] args)
        {
            Scanner sc=new Scanner(System.in);
            System.out.print("Enter no:of elements in the array: ");
            int n=sc.nextInt();
            array a=new array(n);
            System.out.print("Enter number to be searched: ");
            int num=sc.nextInt();
            a.search(num);
        }
    }
```

**<u>Output:</u>**

```
Enter no:of elements in the array: 5
Enter elements to the array:
10 20 30 40 50
Enter number to be searched: 20
20 is found in the array.
```

**<u>Result:</u>** The program has executed successfully and required output is obtained.

Date: 04/02/26

# Program 7: String Manipulation

**Aim:** Java program to demonstrate string manipulation functions.

## Algorithm:

1. Create a class named **word.** Declare instance variable **s** to store the input string.
2. Define a constructor for the **word** class.
   - Prompt the user to enter a string and store it in **s**.
3. Define a method named **str_functions** to perform various string manipulation operations:
   - **s.toLowerCase()**: lowercase version of the string.
   - **s.toUpperCase()**: uppercase version of the string.
   - **s.length():** length of the string;
   - **s.substring(n):** substring of the sting starting from index **n**.
   - **s.substring(n, m):** substring of the string from index **n** to **m.**
   - **s.trim():** trimmed version of the string.
   - **s.indexOf(char):** index of the first occurrence of character **char.**
   - **s.indexOf(char, i):** index of the first occurrence of character **char** starting from index **i.**
   - **s.concat(str2):** concatenate string **str2** with string **s.**
4. In the **main** method:
   - Create an object **w** of the **word** class.
   - Call the **str_functions** method with object **w** to perform the string manipulation operations.

## Source Code:

```
import java.util.*;
class word
{
    Scanner sc=new Scanner(System.in);
    String s;
    word()
    {
        System.out.print("Enter a string: ");
        s=sc.nextLine();
    }
    void str_functions()
    {
        System.out.println("Lower case: "+s.toLowerCase());
        System.out.println("Upper case: "+s.toUpperCase());
        System.out.println("Length: "+s.length());
        System.out.println("substring(2): "+s.substring(2));
        System.out.println("substring(2,6): "+s.substring(2,6));
        System.out.println("trim: "+s.trim());
        System.out.println("indexOf('o'): "+s.indexOf('o'));
```

```
            System.out.println("indexOf('o',10): "+s.indexOf('o',10));
            System.out.println("concat('CR7'): "+s.concat("CR7"));
        }
    }

    class string_manipulation {
        public static void main(String[] args) {
            word w=new word();
            w.str_functions();
        }

    }
```

**Output:**

```
Enter a string:  Cristiano Ronaldo
Lower case:  cristiano ronaldo
Upper case:  CRISTIANO RONALDO
Lenght: 18
substring(2): ristiano Ronaldo
substring(2,6): rist
trim: Cristiano Ronaldo
indexOf('o'): 9
indexOf('o',10): 12
concat('CR7'):  Cristiano RonaldoCR7
```

**Result:** The program has executed successfully and required output is obtained.

Date: 04/02/26

# Program 8: **Array Of Objects**

**Aim:** Program to create a class for Employee having attributes eNo, eName eSalary. Read n employ information and Search for an employee given eNo, using the concept of Array of Objects.

## Algorithm:

1. Create a class named **Employee.** Declare instance variables for employee number (**eNo**), employee name (**eName**), and employee salary (**eSalary**).
2. Define a constructor to read and set **eNo, eName, eSalary.**
3. Prompt the user to enter the number of employees (**n**).
4. Create an array of **Employee** objects (**e[]**) with a size of **n**.
5. Use a **for** loop to iterate through each element of the array:
   - Initialize each element of the array by calling the **Employee** constructor
6. Prompt the user to enter an employee number to search (**num**).
7. Use a **for** loop to search for the employee with the specified number:
   - If found, print the employee name and salary.
   - If not found, print "Employee Not Registered!!!".

## Source Code:

```
import java.util.*;
class Employee
{
    Scanner sc=new Scanner(System.in);
    int eNo;
    String eName;
    double eSalary;
    Employee(int n)
    {
        System.out.println("\t\t\tEnter Details for Employee "+n);
        System.out.print("Enter Employee Number: ");
        eNo=sc.nextInt();
        System.out.print("Enter Employee Name: ");
        eName=sc.next();
        System.out.print("Enter Salary: ");
        eSalary=sc.nextDouble();
    }
}
class array_objects
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter number of employees: ");
```

```java
        int n=sc.nextInt();
        Employee e[]=new Employee[n];
        for(int i=0; i<n; i++)
        {
            e[i]=new Employee(i+1);
        }
        System.out.print("\nEnter Employee Number to search: ");
        int num=sc.nextInt();
        int c=0;
        for(int i=0; i<n; i++)
        {
            if(e[i].eNo==num)
            {
                System.out.println("Employee Name: "+e[i].eName+
                "\nSalary: "+e[i].eSalary);
                c=1;
            }
        }
        if(c==0)
            System.out.println("Employee Not Registered!!!");
    }
}
```

**Output:**

```
Enter number of employees: 3
                        Enter Details for Employee 1
Enter Employee Number: 007
Enter Employee Name: M S Dhoni
Enter Salary: 1234.56
                        Enter Details for Employee 2
Enter Employee Number: 0028
Enter Employee Name: Ronaldo
Enter Salary: 7891.23
                        Enter Details for Employee 3
Enter Employee Number: 0010
Enter Employee Name: Messi
Enter Salary: 4567.89

Enter Employee Number to search: 007
Employee Name: M S Dhoni
Salary: 1234.56
```

**Result:** The program has executed successfully and required output is obtained.

Date: 04/02/26

# Program 9: **Sort Strings**

**Aim:** Write a Java program to sort strings.

## **Algorithm:**

1. Create a class named **sort_string.** Declare instance variables **s[ ]** (array of stringd) and **size.**
2. Define a constructor to read and **size** and **strings.**
3. Prompt the user to enter the number of strings (**size**), and store the stings into the array.
4. Define a method **sort():**
    - Outer loop iterates over each elements of the array.
    - Inner loop iterates over the unsorted portion of the array.
    - Compare adjacent elements of the array and swap then if they are in the wrong order.
    - Repeat the process until the entire array is sorted.
5. Define method **display()** to display the array elements.
6. In the main method, create an object of the **sort_string class.**
7. Display original array.
8. Sort the array using **sort().**
9. Display the sorted string array.

## **Source Code**:

```
import java.util.*;
class sort_string
{
    Scanner sc=new Scanner(System.in);
    String s[];
    int size;
    sort_string()
    {
        System.out.print("Enter no: of strings: ");
        size=sc.nextInt();
        s=new String[size];
        System.out.print("Enter "+size+" strings: ");
        for(int i=0;i<size;i++)
            s[i]=sc.next();
    }
    void sort()
    {
        for(int i=0;i<size;i++)
            for(int j=0;j<size-i-1;j++)
                if (s[j].compareTo(s[j+1])>0)
                {
                    String temp=s[j];
                    s[j]=s[j+1];
```

```
                        s[j+1]=temp;
                    }
        }
        void display()
        {
            for(int i=0;i<size;i++)
                System.out.print(s[i]+" ");
        }
    }


    public class sort_string_main
    {
        public static void main(String[] args)
        {
            sort_string s1=new sort_string();
            System.out.println("\t\tOriginal string array");
            s1.display();
            s1.sort();
            System.out.println("\n\t\t Sorted string array");
            s1.display();
        }
    }
```

**Output:**

```
Enter 3 strings: hello good morning
                Original string array
hello good morning
                 Sorted string array
good hello morning
```

**Result:** The program has executed successfully and required output is obtained.

Date: 11/02/26

# Program 10: **Function Overloading**

**Aim:** Program to calculate area of different shapes using overloaded functions.

## Algorithm:

1. Create a class named **shape_area**, with the following methods.
   - **void area(double r)**: Calculates and prints the area of a circle using the formula: π * r * r.
   - **void area(float r)**: Calculates and prints the area of a square using the formula: side * side.
   - **void area(double l, double b)**: Calculates and prints the area of a rectangle using the formula: length * breadth.
   - **void area(float b, float h)**: Calculates and prints the area of a triangle using the formula: 0.5 * base * height.
2. In the **main** method:
   - Create an object of class **shape_area**.
   - Prompt the user to enter the radius of the circle and call the **area(double r)** method with the provided radius.
   - Prompt the user to enter the length of the side of the square and call the **area(float r)** method with the provided side length.
   - Prompt the user to enter the length and breadth of the rectangle and call the **area(double l, double b)** method with these values.
   - Prompt the user to enter the base and height of the triangle and call the **area(float b, float h)** method with these values.

## Source Code:

```
import java.util.*;
class shape_area
{
    void area(double r)
    {
        System.out.println("Area of circle: "+String.format("%.2f",
         3.14*r*r));
    }
    void area(float r)
    {
        System.out.println("Area of square: "+(r*r));
    }
    void area(double l,double b)
    {
        System.out.println("Area of rectangle: "+(l*b));
    }
    void area(float b,float h)
    {
```

```
            System.out.println("Area of triangle: "+(0.5*b*h));
        }

    }
    class area_main
    {
        public static void main(String[] args)
        {
            Scanner sc=new Scanner(System.in);
            shape_area s =new shape_area();
            System.out.print("Enter radius of circle: ");
            s.area(sc.nextDouble());

            System.out.print("Enter length of side of square: ");
            s.area(sc.nextFloat());

            System.out.print("Enter lenght and breadth of rectangle: ");
            double a=sc.nextDouble();
            double b=sc.nextDouble();
            s.area(a,b);

            System.out.print("Enter breadth and height of triangle: ");
            float a1=sc.nextFloat();
            float b1=sc.nextFloat();
            s.area(a1,b1);
        }
    }
```

**Output:**

```
Enter radius of circle: 3
Area of circle: 28.26
Enter length of side of square: 5
Area of square: 25.0
Enter lenght and breadth of rectangle: 3 6
Area of rectangle: 18.0
Enter breadth and height of triangle: 4 8
Area of triangle: 16.0
```

**Result:** The program has executed successfully and required output is obtained.

Date: 11/02/26

# Program 11: **Single Inheritance**

**Aim:** Create a class 'Employee' with data members Empid, Name, Salary, Address and constructors to initialize the data members. Create another class 'Teacher' that inherit the properties of class employee and contain its own data members department, Subjects taught and constructors to initialize these data members and also include display function to display all the data members. Use array of objects to display details of N teachers.

## Algorithm:

1. Create a class named **Employee.**
   - Declare instance variables **Empid**, **name**, **address**, and **salary.**
   - Define a constructor that initializes the employee details.
2. Define a class **Teacher** inherited from **Employee.**
   - Declare additional instance variable **dept**, **subject[]**, and **no(**number of subjects taught.
   - Define a constructor that initializes teacher details by calling superclass constructors and then initialize the instance variable of class **Teacher.**
   - Define a method **display()** to display the details of the teacher.
3. In the **main** method:
   - Prompt the user to enter number of teachers **(n).**
   - Create an array **t[]** of type **Teacher** with size **n.**
   - Using a loop instantiate each **Teacher** object, prompting the user to enter details for each teacher.
   - Use another loop to display the details of each teacher by calling the **display()** method.

## Source Code:

```
import java.util.*;
class Employee
{
    Scanner sc=new Scanner(System.in);
    int Empid;
    String name,address;
    double salary;
    Employee(int x)
    {
        System.out.println("\n\t\tEnter Details of Teacher "+x);
        System.out.print("Enter Employee Id: ");
        Empid=sc.nextInt();
        System.out.print("Enter Employee Name: ");
        name=sc.next();
        System.out.print("Enter Salary: ");
        salary=sc.nextDouble();
        System.out.print("Enter Address: ");
        address=sc.next();
```

```
    }
}

class Teacher extends Employee
{
    Scanner sc=new Scanner(System.in);
    String dept,subject[];
    int no;
    Teacher(int x)
    {
        super(x);
        System.out.print("Enter Department: ");
        dept=sc.next();
        System.out.print("Enter no:of subjects taught: ");
        no=sc.nextInt();
        subject=new String[no];
        System.err.print("Enter list of subjects: ");
        for(int i=0;i<no;i++)
            subject[i]=sc.next();
    }
    void display(int j)
    {
        System.out.println("\n\t\tTeacher "+j+" Details");
        System.out.println("Employee Id: "+Empid);
        System.out.println("Employee Name: "+name);
        System.out.println("Salary: "+salary);
        System.out.println("Address: "+address);
        System.out.println("Department: "+dept);
        System.out.print("Subjects: ");
        for(int i=0;i<no;i++)
            System.out.print(subject[i]+" ");
    }
}

class teacher_main
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter no:of teachers: ");
        int n=sc.nextInt();
        Teacher t[]= new Teacher[n];
```

```
            for(int i=0;i<n;i++)
                t[i]=new Teacher(i+1);
            for(int i=0;i<n;i++)
                t[i].display(i+1);
        }
    }
```

## Output:

```
Enter no:of teachers: 2

                Enter Details of Teacher 1
Enter Employee Id: 101
Enter Employee Name: ABC
Enter Salary: 25000
Enter Address: House No:123
Enter Department: MCA
Enter no:of subjects taught: 2
Enter list of subjects: OOPS DBMS

                Enter Details of Teacher 2
Enter Employee Id: 102
Enter Employee Name: XYZ
Enter Salary: 20000
Enter Address: House No:897
Enter Department: MCA
Enter no:of subjects taught: 1
Enter list of subjects: NSA

                Teacher 1 Details
Employee Id: 101
Employee Name: ABC
Salary: 25000.0
Address: House
Department: MCA
Subjects: OOPS DBMS
                Teacher 2 Details
Employee Id: 102
Employee Name: XYZ
Salary: 20000.0
Address: House
Department: MCA
Subjects: NSA
```

**Result:** The program has executed successfully and required output is obtained.

Date: 11/02/26

# Program 12: **Multilevel Inheritance**

**Aim:** Create a class 'Person' with data members Name, Gender, Address, Age and a constructor to initialize the data members and another class 'Employee' that inherits the properties of class Person and also contains its own data members like Empid, Company_name, Qualification, Salary and its own constructor. Create another class 'Teacher' that inherits the properties of class Employee and contains its own data members like Subject, Department, Teacherid and also contain constructors and methods to display the data members. Use array of objects to display details of N teachers.

## Algorithm:

1. Create a class named **Person.**
   - Declare instance variables **name, address, gender,** and **age.**
   - Define a constructor to initialize the instance variables.
2. Declare a class **Employee** inherited from **Person.**
   - Declare instance variables **EmpId, CompanyName, qualification,** and **salary.**
   - Define a constructor to call the superclass constructor and initialize employee details.
3. Declare a class **Teacher** inherited from **Employee.**
   - Declare instance variables **subject, department,** and **TeacherID.**
   - Define a constructor to call the superclass constructor and initialize teacher details.
   - Define a **display()** method to display the teacher details.
4. In the **main** method:
   - Prompt the user to enter number of teachers **(n).**
   - Create an array **t[]** of type **Teacher** with size **n.**
   - Using a loop instantiate each **Teacher** object, prompting the user to enter details for each teacher.
   - Use another loop to display the details of each teacher by calling the **display()** method.

## Source Code:

```
import java.util.*;
class Person
{
    Scanner sc=new Scanner(System.in);
    String name,address;
    char gender;
    int age;
    Person(int x)
    {
        System.out.println("\n\n\t\tEnter details of Teacher "+x);
        System.out.print("Enter Name: ");
        name=sc.next();
        System.out.print("Enter Gender(M/F/O): ");
        gender=sc.next().charAt(0);
```

```
        System.out.print("Enter Age: ");
        age=sc.nextInt();
        System.out.print("Enter Address: ");
        address=sc.next();
    }
}

class Employee extends Person
{
    Scanner sc=new Scanner(System.in);
    int EmpId;
    String CompanyName,qualification;
    double salary;
    Employee(int x)
    {
        super(x);
        System.out.print("Enter Employee ID: ");
        EmpId=sc.nextInt();
        System.out.print("Enter Company Name: ");
        CompanyName=sc.next();
        System.out.print("Enter Qualification: ");
        qualification=sc.next();
        System.out.print("Enter Salary: ");
        salary=sc.nextDouble();
    }
}


class Teacher extends Employee
{
    Scanner sc=new Scanner(System.in);
    String subject,department;
    int TeacherId;
    Teacher(int x)
    {
        super(x);
        System.out.print("Enter Teacher ID: ");
        TeacherId=sc.nextInt();
        System.out.print("Enter Subject: ");
        subject=sc.next();
        System.out.print("Enter Department: ");
        department=sc.next();
```

```
        }
        void display(int n)
        {
            System.out.println("\n\n\t\tTeacher "+n+" Details");
            System.out.println("Name: "+name);
            System.out.println("Gender: "+gender);
            System.out.println("Address: "+address);
            System.out.println("Age: "+age);
            System.out.println("Employee ID: "+EmpId);
            System.out.println("Teacher ID: "+name);
            System.out.println("Company Name: "+CompanyName);
            System.out.println("Department: "+name);
            System.out.println("Qualification: "+qualification);
            System.out.println("Salary: "+salary);
            System.out.println("Subject: "+subject);

        }
    }

    class teacher_main
    {
        public static void main(String[] args)
        {
            Scanner sc=new Scanner(System.in);
            System.out.print("Enter no:of teachers: ");
            int n =sc.nextInt();
            Teacher t[]=new Teacher[n];
            for(int i=0;i<n;i++)
                t[i]=new Teacher(i+1);
            for(int i=0;i<n;i++)
                t[i].display(i+1);
        }

    }
```

## Output:

```
Enter no:of teachers: 2


                Enter details of Teacher 1
Enter Name: ABC
Enter Gender(M/F/O): F
Enter Age: 25
Enter Address: House No 256
Enter Employee ID: 101
Enter Company Name: CET
Enter Qualification: PHD
Enter Salary: 35000
Enter Teacher ID: 335
Enter Subject: DBMS
Enter Department: MCA


                Enter details of Teacher 2
Enter Name: XYZ
Enter Gender(M/F/O): M
Enter Age: 26
Enter Address: House No: 876
Enter Employee ID: 102
Enter Company Name: CET
Enter Qualification: PHD
Enter Salary: 35000
Enter Teacher ID: 877
Enter Subject: OOPS
Enter Department: MCA

                Teacher 1 Details
Name: ABC
Gender: F
Address: House
Age: 25
Employee ID: 101
Teacher ID: ABC
Company Name: CET
Department: ABC
Qualification: PHD
Salary: 35000.0
Subject: DBMS


                Teacher 2 Details
Name: XYZ
Gender: M
Address: House
Age: 26
Employee ID: 102
Teacher ID: XYZ
Company Name: CET
Department: XYZ
Qualification: PHD
Salary: 35000.0
Subject: OOPS
```

**Result:** The program has executed successfully and required output is obtained.

Date: 18/02/26

# Program 13: **Interface**

**Aim:** Create an interface having prototypes of functions area() and perimeter(). Create two classes Circle and Rectangle which implements the above interface. Create a menu driven program to find area and perimeter of objects.

## Algorithm:

1. Define the **Shape** interface:
    - Declare two abstract methods **area**() and **perimeter**().
2. Define the **Circle** class:
    - Declare a double variable **radius**.
    - Implement a constructor that takes the **radius** as a parameter and assigns it to the **radius** member variable.
    - Implement the **area**() method to calculate and print the area of the circle.
    - Implement the **perimeter**() method to calculate and print the perimeter of the circle.
3. Define the **Rectangle** class:
    - Declare double variables **length** and **breadth**.
    - Implement a constructor that takes the **length** and **breadth** as parameters and assigns them to the respective member variables.
    - Implement the **area**() method to calculate and print the area of the rectangle.
    - Implement the **perimeter**() method to calculate and print the perimeter of the rectangle.
4. In the **main** method:
    - Declare an integer variable **ch** to store the user's choice.
    - Start a **do-while loop** to display a menu and execute the corresponding choice until the user chooses to exit.
    - Inside the loop, display a menu for the user to choose between Circle, Rectangle, or Exit.
    - If the user chooses **Circle**:
        - Prompt the user to enter the **radius** of the circle.
        - Create a Circle object with the given radius.
        - Call the **area**() and **perimeter**() methods of the Circle object.
    - If the user chooses Rectangle:
        - Prompt the user to enter the **length** and **breadth** of the rectangle.
        - Create a Rectangle object with the given length and breadth.
        - Call the **area**() and **perimeter**() methods of the Rectangle object.
    - If the user chooses Exit:
        - Print "Exiting..." and exit the loop.
    - If the user enters an **invalid choice**, display "Invalid choice!".
    - Repeat the loop until the user chooses to exit (option 3).

## Source Code:

```
import java.util.*;
interface Shape
{
```

```
    public void area();
    public void perimeter();
}


class Circle implements Shape
{
    double radius;
    Circle(double r)
    {
        radius = r;
    }
    public void area()
    {
        System.out.println("Area of circle: " + String.format("%.2f",
       (3.14*radius*radius)));
    }
    public void perimeter()
    {
        System.out.println("Perimeter of circle: " + String.format(
       "%.2f",(2*3.14*radius)));
    }
}


class Rectangle implements Shape
{
    double length,breadth;
    Rectangle(double l,double b)
    {
        length = l;
        breadth = b;
    }
    public void area()
    {
        System.out.println("Area of rectangle: " + String.format("%.2f"
       ,(length*breadth)));
    }
    public void perimeter()
    {
        System.out.println("Perimeter of rectangle: " + String.format(
       "%.2f",(2*(length+breadth))));
```

```
    }
}



class circle_rect {
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int ch;
        do
        {
            System.out.print("------------------\n1.Circle\n2.Rectangle
            \n3.Exit\n------------------\nEnter your choice: ");
            ch=sc.nextInt();
            switch(ch)
            {
                case 1:
                        System.out.print("Enter radius of circle: ");
                        double r=sc.nextDouble();
                        Circle c =new Circle(r);
                        c.area();
                        c.perimeter();
                        break;
                case 2:
                        System.out.print("Enter length and breadth of
                        rectangle: ");
                        double l=sc.nextDouble();
                        double b=sc.nextDouble();
                        Rectangle r1 =new Rectangle(l,b);
                        r1.area();
                        r1.perimeter();
                        break;
                case 3:
                        System.out.println("Exiting...");
                        break;
                default:
                        System.out.println("Invalid choice!");
            }
        }while(ch!=3);
    }
}
```

## Output:

```
------------------
1.Circle
2.Rectangle
3.Exit
------------------
Enter your choice: 1
Enter radius of circle: 4
Area of circle: 50.24
Perimeter of circle: 25.12
------------------
1.Circle
2.Rectangle
3.Exit
------------------
Enter your choice: 2
Enter length and breadth of rectangle: 3 7
Area of rectangle: 21.00
Perimeter of rectangle: 20.00
------------------
1.Circle
2.Rectangle
3.Exit
------------------
Enter your choice: 3
Exiting...
```

**Result:** The program has executed successfully and required output is obtained.

Date: 18/02/26

# Program 14: **Package**

**Aim:** Create a Graphics package that has classes and interfaces for figures Rectangle, Triangle, Square and Circle. Test the package by finding the area of these figures.

**Algorithm:**

1. Define the **Figure** interface, and declare an abstract method **area**().
2. Define the **Circle** class:
   - Declare a double variable **radius**.
   - Implement a **constructor** that takes the **radius** as a parameter and assigns it to the **radius** member variable.
   - Implement the **area**() method to calculate and print the area of the circle.
3. Define the **Rectangle** class:
   - Declare double variables **length** and **breadth**.
   - Implement a **constructor** that takes the **length** and **breadth** as parameters and assigns them to the respective member variables.
   - Implement the **area**() method to calculate and print the area of the rectangle.
4. Define the **Square** class:
   - Declare a double variable **length**.
   - Implement a constructor that takes the **length** as a parameter and assigns it to the length member variable.
   - Implement the **area**() method to calculate and print the area of the square.
5. Define the **Triangle** class:
   - Declare double variables **breadth** and **height**.
   - Implement a constructor that takes the **breadth** and **height** as parameters and assigns them to the respective member variables.
   - Implement the **area**() method to calculate and print the area of the triangle.
6. In the main method:
   - Prompt the user to enter the **radius** of a **circle** and read the input.
   - Create a **Circle object** with the given radius.
   - Call the **area**() method of the Circle object to calculate and print the area of the circle.
   - Prompt the user to enter the **length** and **breadth** of a rectangle and read the inputs.
   - Create a **Rectangle object** with the given length and breadth.
   - Call the **area**() method of the Rectangle object to calculate and print the area of the rectangle.
   - Prompt the user to enter the **side of a square** and read the input.
   - Create a Square object with the given side length.
   - Call the **area**() method of the Square object to calculate and print the area of the square.
   - Prompt the user to enter the **height** and **breadth** of a triangle and read the inputs.
   - Create a Triangle object with the given height and breadth.
   - Call the **area**() method of the Triangle object to calculate and print the area of the triangle.

**Source Code**:

```
package Graphics;

interface Figure
{
    public void area();
}

public class Circle implements Figure
{
    double radius;
    public Circle(double r)
    {
        radius = r;
    }
    public void area()
    {
        System.out.println("Area of circle: " + String.format("%.2f"
         ,(3.14*radius*radius)));
    }

}

public class Rectangle implements Figure
{
    double length;
    double breadth;
    public Rectangle(double l,double b)
    {
        length=l;
        breadth=b;
    }
    public void area()
    {
        System.out.println("Area of Rectangle: " + String.format("%.2f"
         ,(length*breadth)));
    }
}

public class Square implements Figure
{
```

```
    double length;
    public Square(double l)
    {
        length=l;
    }
    public void area()
    {
        System.out.println("Area of Square: " + String.format("%.2f"
         ,(length*length)));
    }
}

public class Triangle implements Figure
{
    double breadth;
    double height;
    public Triangle(double b,double h)
    {
        height=h;
        breadth=b;
    }
    public void area()
    {
        System.out.println("Area of Triangle: " + String.format("%.2f"
         ,(0.5*height*breadth)));
    }
}
```

## Class Using Graphics Package

```
import java.util.Scanner;
import Graphics.*;

class Test_package {
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.print("\nEnter radius of circle: ");
        double cr=sc.nextDouble();
        Graphics.Circle c = new Graphics.Circle(cr);
        c.area();
```

```
        System.out.print("\nEnter length and breadth of rectangle: ");
        double l=sc.nextDouble();
        double b=sc.nextDouble();
        Graphics.Rectangle r = new Graphics.Rectangle(l,b);
        r.area();
        System.out.print("\nEnter side of square: ");
        l=sc.nextDouble();
        Graphics.Square s = new Graphics.Square(l);
        s.area();
        System.out.print("\nEnter height and breadth of triangle: ");
        l=sc.nextDouble();
        b=sc.nextDouble();
        Graphics.Triangle t = new Graphics.Triangle(b,l);
        t.area();
    }
}
```

**Output:**

```
Enter radius of circle: 3
Area of circle: 28.26

Enter length and breadth of rectangle: 4 7
Area of Rectangle: 28.00

Enter side of square: 5
Area of Square: 25.00

Enter height and breadth of triangle: 3 7
Area of Triangle: 10.50
```

**Result:** The program has executed successfully and required output is obtained.